



Hederagenin and α -hederin promote degradation of proteins in neurodegenerative diseases and improve motor deficits in MPTP-mice

An-Guo Wu^{a,1}, Wu Zeng^{a,1}, Vincent Kam-Wai Wong^a, Yi-Zhun Zhu^b, Amy C.Y. Lo^c,
Liang Liu^{a,*}, Betty Yuen-Kwan Law^{a,*}

^a State Key Laboratory of Quality Research in Chinese Medicine, Macau University of Science and Technology, Macau, China

^b School of Pharmacy, Macau University of Science and Technology, Macau, China

^c Department of Ophthalmology, Li Ka Shing Faculty of Medicine, The University of Hong Kong, Hong Kong Special Administrative Region, China

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ABSTRACT

Pathogenesis of neurodegenerative diseases such as Parkinson's disease (PD) and Huntington's disease (HD) are closely related to the formation of protein aggregates and inclusion body. For instance, active autophagic components from Chinese herbal medicines (CHMs) are highlighted to modulate neurodegeneration via degradation of disease proteins. In this study, the neuroprotective effect of the purified *Hedera helix* (HH) fraction containing both hederagenin and α -hederin, is confirmed by the improvement of motor deficits in PD mice model. Furthermore, hederagenin and α -hederin derived from HH are confirmed as novel autophagic enhancers. Both compounds reduce the protein level of mutant huntingtin with 74 CAG repeats and A53T α -synuclein, and inhibit the oligomerization of α -synuclein and inclusion formation of huntingtin, via AMPK-mTOR dependent autophagy induction. Both hederagenin and α -hederin induce autophagy and promote the degradation of neurodegenerative mutant disease proteins in vitro, suggesting the therapeutic roles of HH in neurodegenerative disorders.

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1. Introduction

Hedera helix (HH) is an ornamental plant, which is prescribed traditionally for modulating respiratory tract diseases with severe mucous formation, infections, or cough resulting from common cold, through exhibiting its anti-mutagenic [1] and anti-inflammatory bioactivities [2]. Triterpenoid saponins are the major components [3,4] responsible for the biological activities of HH [5–8]. While most of the saponins in HH are hederagenin-type triterpene saponins, hederagenin was identified as the bioactive chemical component in HH [9], which possessed anti-depressant

[10] and anti-inflammatory effects [11] in rodent models. α -hederin is a secondary saponin isolated from HH [12]. Its biological effects are mainly on anti-cancer [13,14], anti-diabetes [15], anti-inflammation [16] and anti-oxidation [8].

Autophagy is an essential intracellular degradation system that delivers unwanted cytoplasmic constituents to lysosome for recycling of cellular building blocks. This cellular degradation process starts with the sequential formation, elongation, and fusion of an isolated membrane to form double membrane vesicles (autophagosomes) that sequester unwanted cytoplasmic materials [17,18]. The role of autophagy in neurodegenerative disorders has been intensively studied [19,20]. It was proposed that autophagy plays a pivotal role in neuroprotection through facilitating the degradation of aggregate-prone and misfolded disease proteins, which are highly correlated to the onset of neurodegenerative diseases (proteinopathies) [21]. These include Alzheimer's disease (AD), Parkinson's disease (PD), tauopathies, and polyglutamine expansion diseases such as Huntington's disease (HD) and spinocerebellar ataxias [18,22]. It was reported that mutations lead to proteinopathies by conferring new toxic functions on specific diseases, whose expression levels are highly correlated to neurodegeneration [23]. Therefore, facilitating the clearance of aggregate-prone disease proteins via autophagic degradation is putative therapeutic approach in neuro-therapy.

Abbreviations: CHMs, Chinese herbal medicines; ATG, autophagy-related proteins; AD, Alzheimer's disease; PD, Parkinson's disease; AMPK, 5' adenosine monophosphate-activated protein kinase; mTOR, mammalian target of rapamycin; HD, Huntington's disease; HH, *hedera helix*; PF, petroleum; MPTP, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine; EF, ethylacetate fraction; HTT, huntingtin; NF, *n*-butanol; GFP, green fluorescent protein; PE, phosphatidylethanolamine; AHS, acid hydrolyzed solution; CMC, cell membrane chromatography; 3-MA, 3-methyladenine; TSC2, tuberous sclerosis complex; CC, compound C; polyQ tract, polyglutamine tract; MEF, mouse embryonic fibroblasts; Dox, doxycycline; α -syn, α -synuclein; BiFC, bimolecular fluorescence complementation (BiFC) model; MAOB, monoamine oxidase B; DA, dopamine.

* Corresponding authors.

E-mail addresses: lliu@must.edu.mo (L. Liu), yklaw@must.edu.mo (B.Y.-K. Law).

¹ Equal contributors.

HD is an autosomal-dominant neurodegenerative disease caused by excess (>36) number of CAG trinucleotide repeat expansion, which results in a long mutant polyQ tract in the mutant huntingtin protein [24]. These repeated expansions are associated with the formation of aggregate-prone or mutant proteins, which lead to cytotoxicity in cellular or *in vivo* models [25,26]. The clearance of the mutant huntingtin is highly dependent on autophagy for degradation. PD is associated with accumulation of protein aggregates, or α -synuclein which is one of the major components of Lewy body. A53T and A30P α -synuclein mutants were proved to be a substrate of autophagy, therefore it was proposed that autophagy plays an important role in neuro-protection via the cellular degradation of mutant proteins [27].

Currently, more and more researches are focusing on the discovery of novel small molecules for autophagy induction. Besides, many chemical components or extracts from Chinese herbal medicines (CHM) were identified to induce autophagy and facilitate clearance of the aggregated proteins in different neurodegenerative diseases [24,28,29]. In this study, through screening of our compound library of CHM, we found that the ethylacetate fraction (EF) of the total ethanol extract of HH can induce autophagy and exhibit the neuroprotective effect by improving the locomotion of neuro-toxin (MPTP)-injected mice with induced motor deficits. In addition, we have identified two major components, hederagenin and α -hederin in the HH-EF, as autophagy inducers. These two single compounds can activate autophagy via the AMPK-mTOR and autophagy-related gene (ATG) 7 signaling pathways. Furthermore, the two compounds were able to accelerate the clearance of mutant huntingtin (HTT) with 74 CAG repeats, A53T α -synuclein, and inhibit the oligomerization of α -synuclein in cellular models.

2. Results

2.1. Fractionation of crude extract of HH to identify novel autophagic activators

To identify potential active autophagic compounds from HH, the whole plant was smashed into fine power and extracted with 75% ethanol. Dried ethanol HH extract was re-dissolved in water, and then successively partitioned with petroleum ether (PF), ethylethanoate (EF) and *n*-butanol (NF) to give the petroleum ether HH fraction (HH-PF), ethylethanoate HH fraction (HH-EF), *n*-butanol HH fraction (HH-NF) and water HH fraction (HH-WF), respectively. To begin, the IC₅₀ value of all the isolated fractions were evaluated using MTT assay (Fig. 1a). Pheochromocytoma-derived PC-12 cells have been commonly adopted for the cellular study of neurotrophin action, protein trafficking or neural differentiation [30,31]. PC-12 cells have also been used for the study of dopaminergic neurons and PD as they display metabolic features of PD [32]. We therefore studied the autophagic effect of all the isolated HH fractions in green fluorescent protein (GFP)-LC3 transfected PC-12 cells [33]. During autophagy induction, LC3-I is activated by ATG genes, followed by the conjugation of LC3-I to phosphatidylethanolamine (PE), then cytosolic LC3-I become membrane-bound LC3-II for the formation of autophagosome [34]. Through observing the conversion of LC3 by immunofluorescence microscopy, Fig. 1b showed the EF fraction of HH increases the formation of fluorescent LC3-II puncta in PC-12 cells. This result was further confirmed by Western blot analysis (Fig. 1c) which confirmed an increase in the protein level of LC3-II in PC-12 cells.

2.2. The saponin and the saponins with low polarity induce autophagy

According to the reported literatures, saponins are the major chemical components in HH, which possess antispasmodic,

antileishmanial, antifungal, anthelmintic, molluscicidal and antimutagenic properties [35]. In general, while saponins containing ester glycosides (R2 is Glycosylation) are with higher polarity that can be extracted by *n*-butanol, saponins containing the oxygen glycosides (R1 is Glycosylation or H) have relatively low polarity that are extracted with ethylacetate (Fig. 2a) [35,36].

As shown in previous section, the EF fraction of HH possesses the highest autophagic effect, therefore, we hypothesized that both saponin and saponins with low polarity may be responsible for inducing autophagy in PC-12 cells. To this end, the NF fraction of HH (HH-NF) was acidically hydrolyzed by adding hydrochloric acid to obtain a solution of pH 2.5. The solution was then heated to completely hydrolyze all the saponins (ester glycosides). The resulting acid hydrolyzed solution (AHS) was extracted with ethylacetate (Fig. 2a). We then evaluated the cytotoxicity value (IC₅₀) and autophagic effect of both HH-NF(AHS)-WF and HH-NF-(AHS)-EF fractions in PC-12 cells.

As shown in Fig. 2b, while both HH-NF(AHS)-WF and HH-NF(AHS)-EF possesses similar IC₅₀ values of 42.6 and 57.4 μ g/mL, respectively, HH-NF(AHS)-EF induces potent autophagic effect as revealed by high percentage of cells with GFP-LC3 puncta formation (Fig. 2c). This result was further confirmed by Western blot which showed an increased expression of LC3-II in cells after treatment of HH-NF(AHS)-EF (Fig. 2d). Consistent with the result that HH-NF(AHS)-WF did not induce autophagy (Fig. 2c and d), we postulated that the saponin and saponins with low polarity in both HH (EF) and HH-NF(AHS)-EF may be responsible for the autophagic effect of HH.

2.3. HH improves motor deficits of neurotoxin (MPTP)-induced PD mice models

Up to now, classical cellular and animal models set up by the addition of neurotoxin MPTP, are the most widely used PD platforms for testing neuroprotective drugs [37]. These models are well established to induce motor deficits and neurotoxicity in *drosophila*, rats, mice, and primates [38–40]. MPTP, a highly lipophilic compound which crosses the blood brain barrier, are converted by astrocytes to its toxic 1-methyl-4-phenylpyridinium (MPP⁺) metabolite ion, and resembles a number of known environmental neurotoxins such as herbicide (paraquat) [41] or insecticide (rotenone) [42], to induce degeneration of dopaminergic neuron.

Motor dysfunctions of PD include rigidity, tremor and akinesia [43]. Recent study in the use of swim test in monitoring the behavioral deficits of MPTP-induced PD mice models, demonstrated a negative correlation between swim-score and MPTP level in mice, and confirming the use of swim test as a reliable technique to motor deficits of MPTP-induced models [44]. To evaluate the protective function of HH in motor dysfunctions repairment of MPTP-treated mice, HH (EF) and HH-NF(AHS)-EF in 1:1 ratio were administrated to (10 days of pre-treatment) and after the start of MPTP treatment for 10 consecutive days. To begin, C57BL/6 mice were treated with 20 mg/kg of MPTP for 10 consecutive days, HH (EF) and HH-NF(AHS)-EF in 1:1 ratio was administrated daily to mice 2 h before MPTP injection. Mice were subjected to swim-test for every 3 days after the start of MPTP injection. As shown in Fig. 3a, while the control C57BL/6 mice demonstrated a better swimming ability as indicated by a higher swim score, MPTP-treated mice showed a relatively lower swim score. As compared to the group treated with MPTP, HH extract-treated group showed a significant recovery in swim score, suggesting the protective role of HH in the repairment of motor deficits induced by MPTP.

Another common way to evaluate motor deficit of MPTP-induced mice is by the rotarod test. This is performed by measuring the duration the mice ran on the rotating rod by using an automated device. To begin, motor coordination of mice was assessed

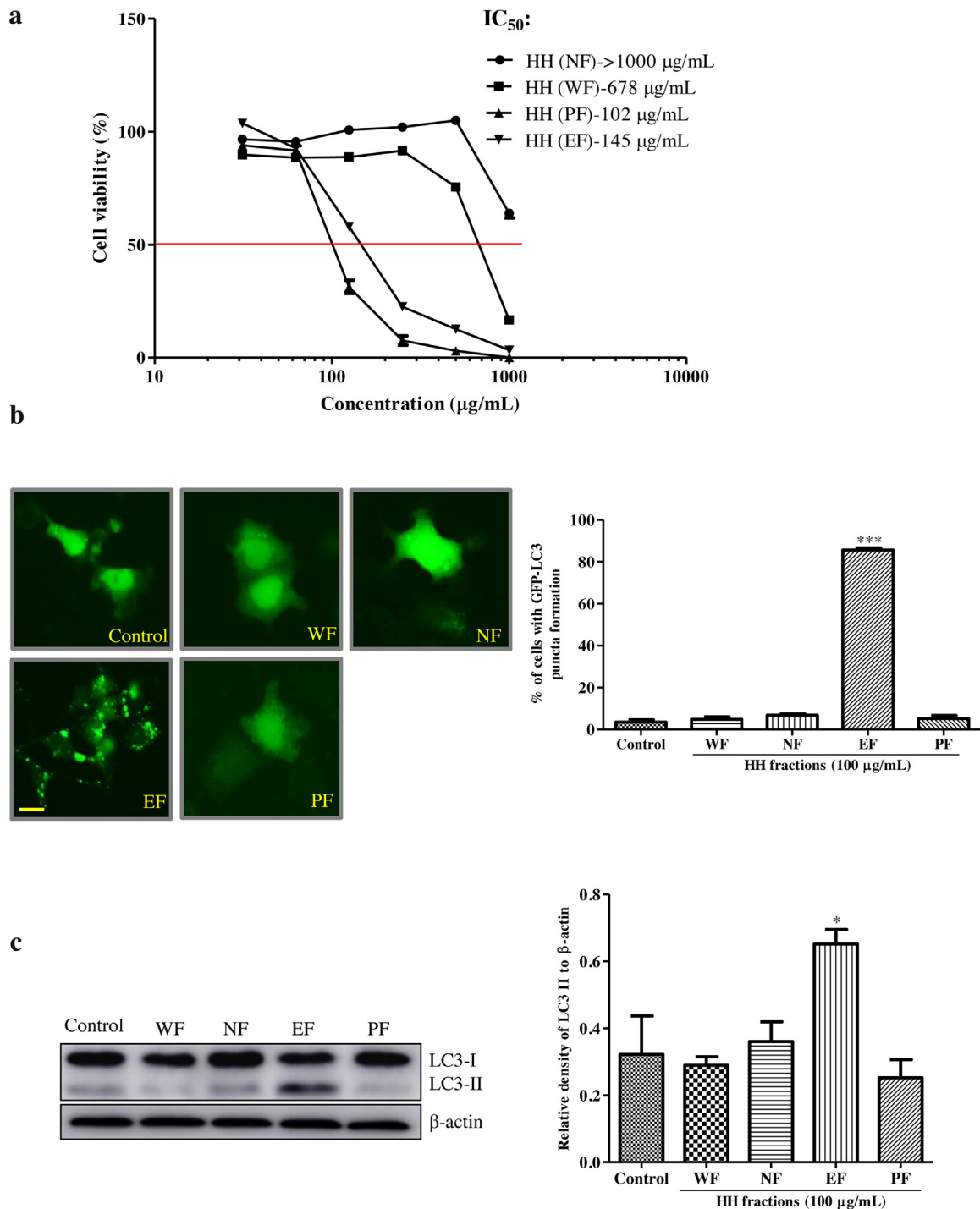


Fig. 1. Cytotoxicity and autophagic effect of the HH fractions extracted by NF, WF, PF or EF.

(a) IC₅₀ values of the NF, WF, PF and EF fractions of HH were evaluated by MTT assay after 48 h of treatments. (b) PC-12 cells transfected with GFP-LC3 plasmids were incubated with different HH fractions (100 $\mu\text{g/mL}$) as indicated for 24 h. Representative fluorescence images with the formation of GFP-LC3 puncta were shown. Right: percentage of cells with GFP-LC3 puncta formation. (c) PC-12 cells treated with different fractions of HH (100 $\mu\text{g/mL}$, 24 h) were harvested for analysis of LC3 I/II and β -actin protein level. Band intensities were quantified using densitometric analysis with normalization to β -actin. Bars are representatives of three independent experiments. Error bars, S.D. *** $p < 0.001$; * $p < 0.05$. Magnification, $\times 40$; Scale bar: 15 μm . The full-length blots are presented in Supplementary Fig. S4.

by placing each C57BL/6 mouse on the horizontal rod that keeps rotating about its axis [45]. The time that the mouse remains upright and running without falling off were recorded. As shown in Fig. 3b, while the MPTP-treated mice demonstrated a significant shorter duration in maintaining upright on the rotating rod without falling, HH extract-treated mice showed a significant longer duration remaining on the rod as measured at day 1, day 4 and day 10

post-MPTP injections, implying the improvement of mice motor function after HH treatments. Further analysis by UHPLC-MS/MS system showed that both hederagenin (3.39 nM) and α -hederin (2.98 nM) could be detected in the brain tissue (with 80 mg/kg HH treatment) (Fig. 3c), suggesting that these two potential neuroprotective agents are able to cross the blood-brain barrier, which may be responsible for the protective effect of HH.

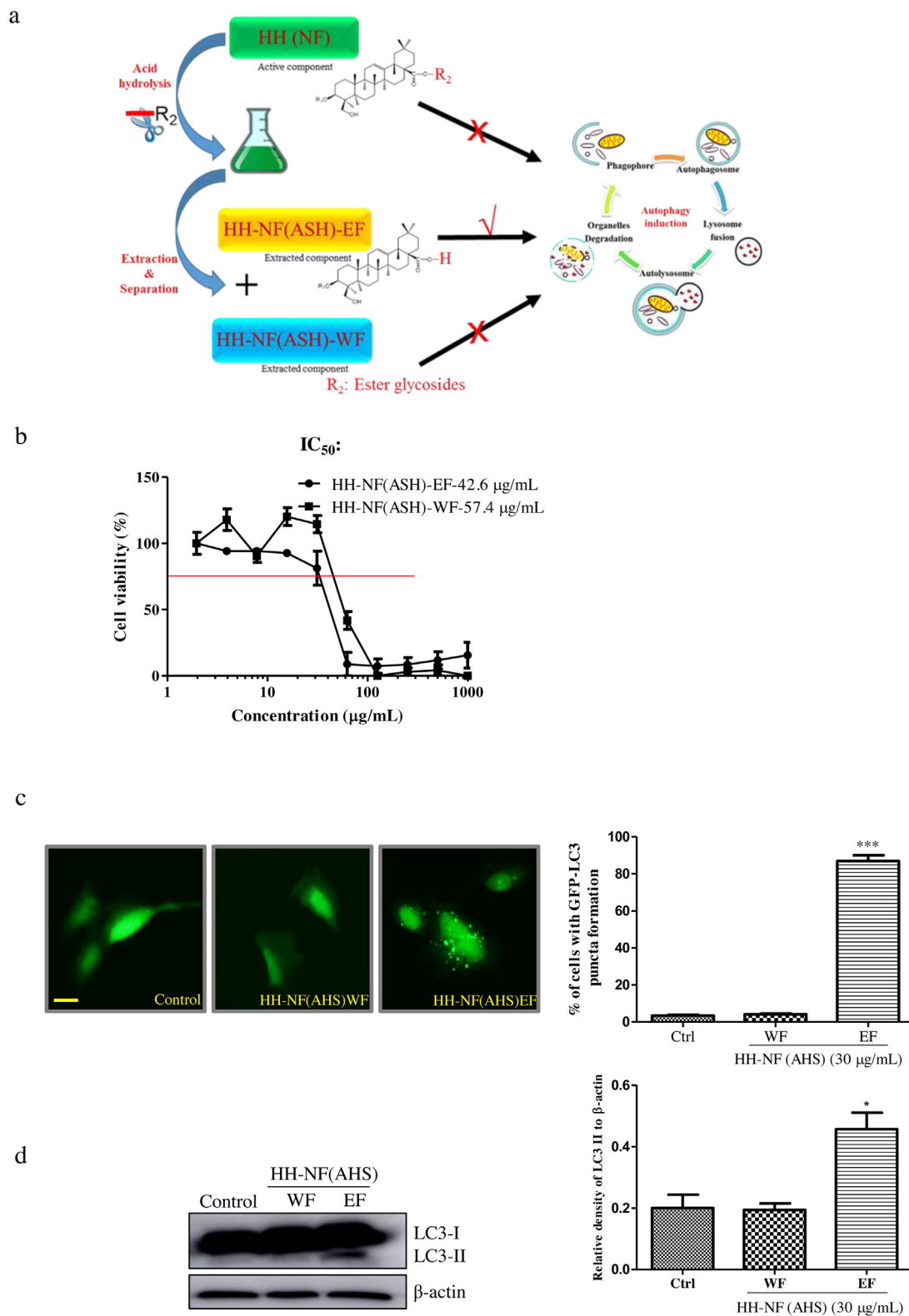


Fig. 2. The autophagic effect of the hydrolyzed fractions (ASH) of HH (NF).

(a) Schematic diagram showing the resulted fractions after acid hydrolysis of NF fraction of HH. It is hypothesized that saponins with high polarity are extracted into the *n*-butanol fraction, and can be hydrolyzed into sapogenin or saponins with lower polarity. The final hydrolyzed fraction was then extracted with ethylacetate to give the HH-NF(ASH)-WF and HH-NF(ASH)-EF, respectively. The autophagic role of each fraction was then evaluated. (b) Cytotoxicity of both HH-NF(ASH)-WF or HH-NF(ASH)-EF in PC-12 cells after 48 h of treatment. (c) PC-12 cells transfected with GFP-LC3 plasmids were treated with either HH-NF(ASH)WF or HH-NF(ASH)EF (30 μ g/mL) for 24 h. Representative images showed the formation of GFP-LC3 puncta. The percentage of cells with GFP-LC3 puncta formation was showed in the bar chart. (d) PC-12 cells were treated with different fractions of HH (ASH) (30 μ g/mL) for 24 h as indicated. Cell lysates were harvested for analyzing the protein level of LC3 I/II and β -actin. Right: LC3-II band intensities were quantified and normalized to β -actin. Bars are representatives of three independent experiments. Error bars, S.D. *** p < 0.001; * p < 0.05. Magnification, $\times 40$; Scale bar: 15 μ m. The full-length blots are presented in Supplementary Fig. S5.

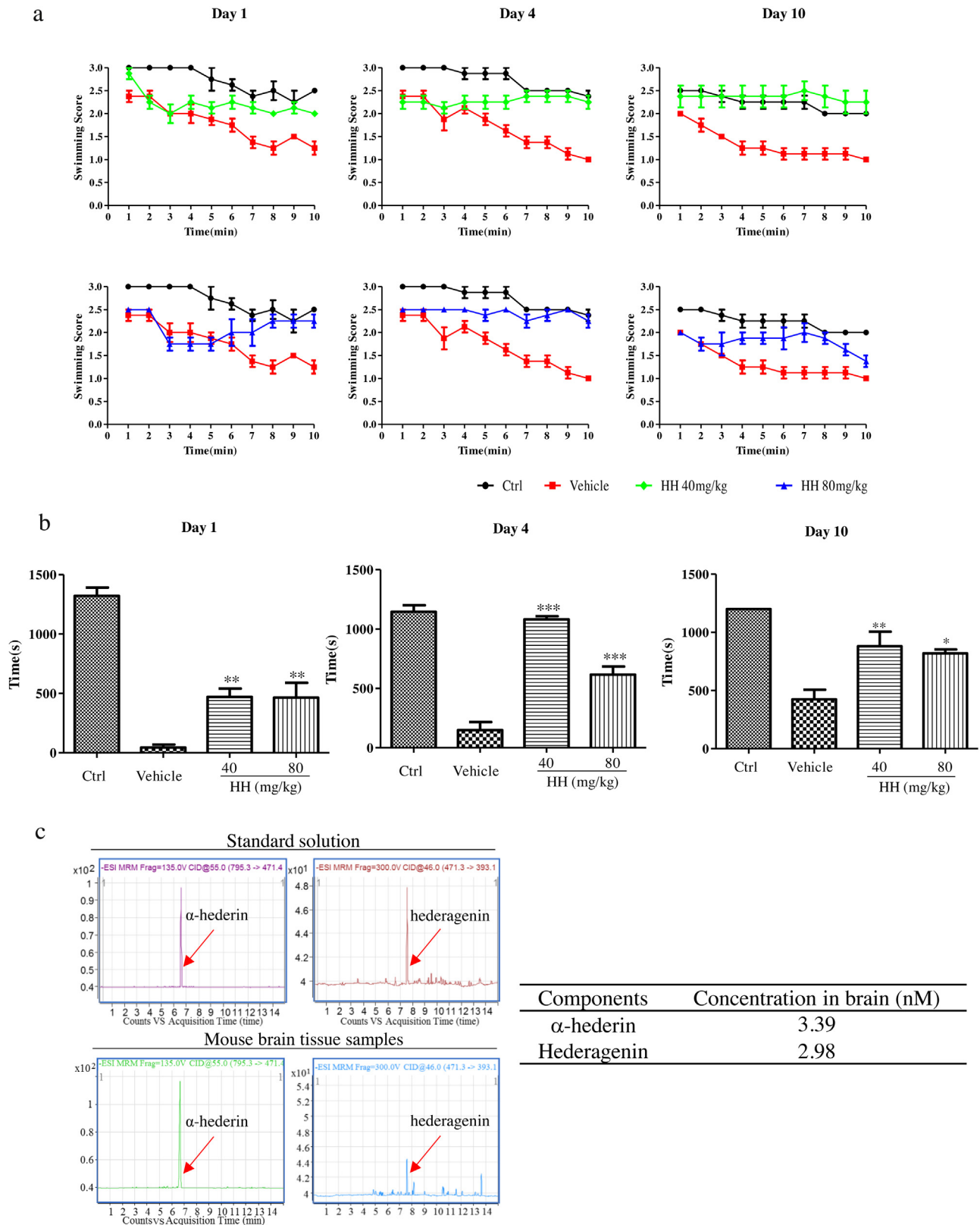


Fig. 3. Mixture of HH (EF) and HH-NF (ASH)-EF improves motor deficits of C57 mice injected with MPTP.

(a) Effect of MPTP on the swimming ability of normal control or MPTP-injected mice (with or without HH extract treatments) was scored at day 1, day 4 and day 10 post-MPTP treatments. At 30 min after MPTP injection (20 mg/kg/day, i.p.), swim-scores were evaluated for a total of 10 min with 1 min interval. Each point represented the mean swimming score of 8 animals $\pm$ S.E.M. (b) Rotarod tests were repeated for 3 times in each individual mouse to give a mean score in second(s). Scores were taken at day 1, day 4 and day 10 post-MPTP treatments (with or without HH extract treatments). Each column represents the mean $\pm$ S.E.M. value of 8 mice in each individual treatment group. (c) The measurement of hederagenin and α -hederin in mouse brain was carried out by using ultra-high performance liquid chromatography-tandem mass spectrometry (UHPLC-MS/MS) with multiple reactions monitoring (MRM) mode. Representative MRM ion chromatograms of hederagenin and α -hederin in standard solution and mouse brain samples were shown. The concentration of hederagenin and α -hederin in mouse brain samples (treatment group: 80 mg/kg HH) were shown in the table. Control: no treatment; vehicle: MPTP-injected control. Bars, S.E.M. *** p < 0.001; ** p < 0.01; * p < 0.05.

2.4. The identification of the active autophagic components in HH (EF) and HH (AHS)-EF

Through analyzing HH (NF), HH-NF(AHS)-EF and HH (EF) by UHPLC-TOF/MS, chromatographic peaks confirmed that the compounds (saponins), eluted between 4 and 6 min of HH (NF) separation, were hydrolyzed into the compounds eluted between 6.5–9.5 min of HH-NF(AHS)-EF separation. Concordantly, chromatographic peak pattern of HH-NF(AHS)-EF was similar as the peaks eluted between 6.5 to 9.5 min of HH (EF) separation (Fig. 4a), suggesting the major chemical compounds presented in the HH-NF(AHS)-EF are similar to HH (EF), which may be responsible for the induction of autophagy.

In order to identify the active components responsible for the autophagic effect of HH (EF), we applied the cell membrane chromatography (CMC) to identify the chemical components that possess binding affinity to the cellular membrane of PC-12 cells [33]. This is done by incubating PC-12 cells with the HH (EF) for 4 h. After incubation, while chemical components without binding affinity to the cell membrane were washed away by PBS buffer, those components that could bind on cell membrane were retained for analysis. As shown in Fig. 4b, the components that were bound on the cellular membrane were showed in brown line of the chromatogram, while the chromatograms of blank lysate (black line) or PBS buffer (green line) were showed as control. By comparing the accurate mass and molecular formula of the identified compounds from CMC to the reported compounds of HH in the literatures, hederagenin and α -hederin were identified as the possible components responsible for the autophagic activities of HH (Fig. 4c).

In addition, the concentration and the percentage of both hederagenin and α -hederin in HH-NF(AHS)-EF, HH (NF) and HH (EF) were measured. As shown in Fig. 4d, only a small amount of hederagenin (0.42 μ M) and α -hederin (0.38 μ M) are presented in HH (NF) when compared to HH-NF(AHS)-EF or HH (EF). Furthermore, HH-NF(AHS)-EF contained a higher concentration of hederagenin (74.42 μ M) and α -hederin (22.62 μ M) when compared to HH (EF) (2.95 μ M and 19.88 μ M respectively). Therefore, we suggested that hederagenin and α -hederin may work as the autophagy inducers responsible for the bioactivity of HH.

2.5. Hederagenin and α -hederin increase LC3-II level in PC-12 cells

Firstly, the IC_{50} value of both hederagenin and α -hederin was evaluated by MTT assay for 48 h. The mean IC_{50} value of hederagenin and α -hederin was estimated as >100 μ M and 26.5 μ M, respectively (Fig. 5a). Furthermore, both hederagenin and α -hederin trigger an increase in the protein level of LC3-II via a dose- and time-dependent manner (Fig. 5b). The autophagic effect of hederagenin and α -hederin was further validated by monitoring and quantitating the formation of GFP-LC3-II puncta by immunofluorescence microscopy. As shown by an increased formation of fluorescent LC3-II puncta in cells (Fig. 5c and d), we confirmed that both hederagenin and α -hederin trigger the induction of autophagy (time- and dose-dependently) in PC-12 cells.

2.6. Hederagenin and α -hederin induce autophagic flux in PC-12 cells

To confirm the autophagic flux induced by hederagenin and α -hederin, lysosomal degradation was inhibited by the addition of two lysosomal protease inhibitors, E64d and pepstatin A, when evaluating the protein level of the key autophagic protein, LC3-II. This is because the failure in fusion of autophagosomes and lysosomes could also lead to an increased protein level of LC3-II in Western blot, or GFP-LC3 puncta formation in fluorescence images,

due to a reduction in turnover of autophagosome [17]. As shown in Fig. 6a, both hederagenin and α -hederin significantly increase the rate of LC3-II formation with the presence of E64d and pepstatin A, when compared with treatment of compounds alone. This result confirmed the dose-dependent induction of autophagic flux after hederagenin and α -hederin treatments. To further confirm the compounds-mediated autophagy flux, 3-methyladenine (3-MA) which blocks autophagy via inhibition of type III phosphatidylinositol 3-kinases was used. Consistently, the addition of 3-MA decreases the rate of autophagy as demonstrated by a decreased protein level of LC3-II (Fig. 6b) and percentage of cells with GFP-LC3 puncta formation (Fig. 6c) after hederagenin or α -hederin treatments, suggesting the autophagic property of the two compounds from HH.

2.7. Hederagenin and α -hederin induce autophagy via AMPK-mTOR signalling cascade

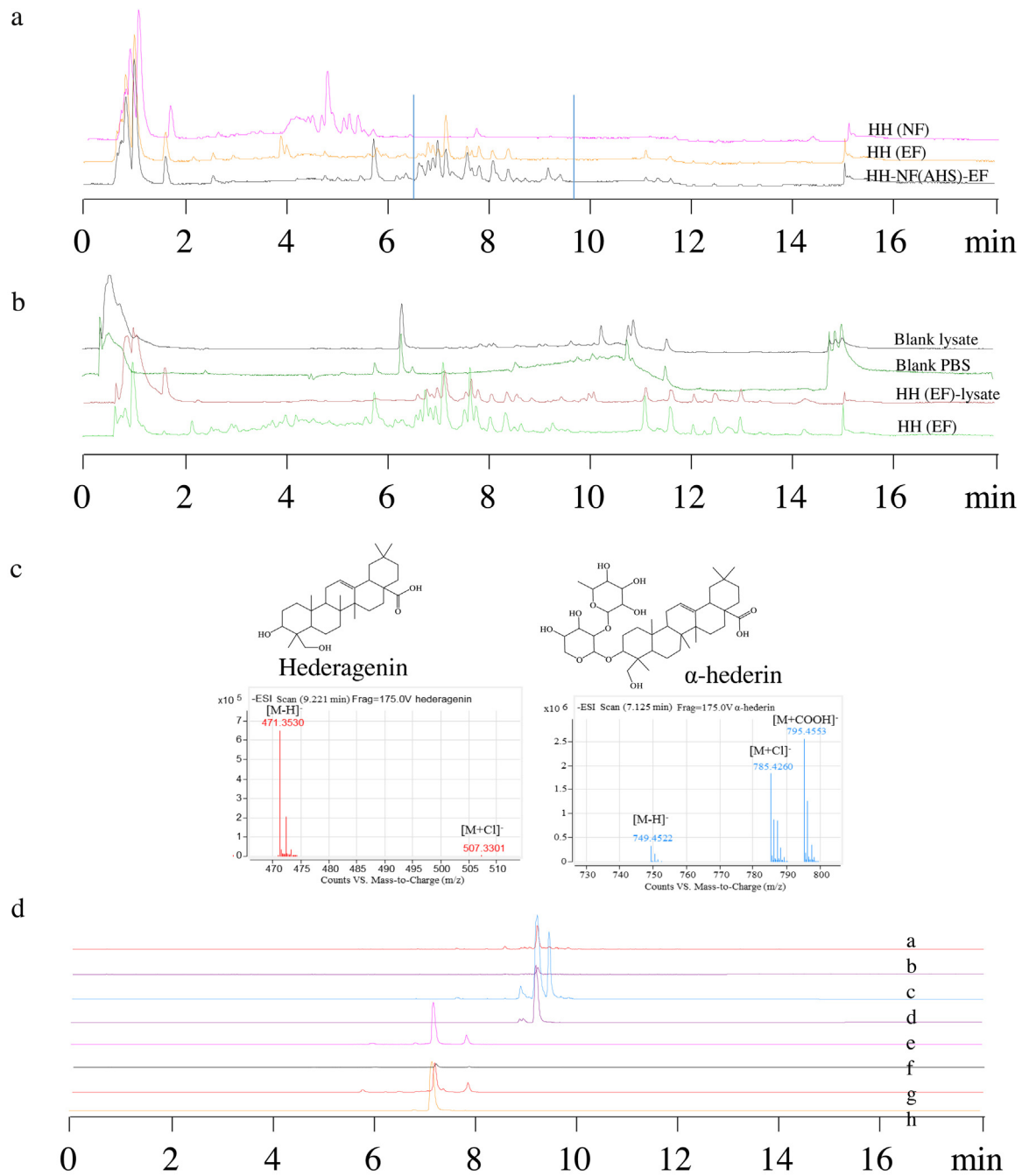
Autophagy is promoted by the activation of AMP activated protein kinase (AMPK), a key energy sensor for maintaining normal cellular metabolism and energy homeostasis, especially under low intracellular ATP conditions such as starvation or hypoxia. To further explore the possible mechanistic action of hederagenin or α -hederin, the effect on the phosphorylation of AMPK was therefore investigated [46]. As shown in Fig. 7a, either hederagenin or α -hederin induce an increased phosphorylation of AMPK.

Downstream of AMPK activation is the phosphorylation of the tumor suppressing protein tuberous sclerosis complex (TSC2), and the suppression of mTOR which works as the central regulator for well-coordination of growth factors and signals for cell growth [47]. While the suppression of mTOR could lead to the induction of autophagy, hederagenin or α -hederin lead to a reduced phosphorylation of p70S6K, which is a downstream target of mTOR (Fig. 7a). Further investigation on the AMPK pathway was studied by the additional of AMPK inhibitor compound C (CC) during hederagenin and α -hederin treatments. As shown in Fig. 7b, there is a significant reduction in protein level of LC3-II and the formation of GFP-LC3 puncta (Fig. 7c) in hederagenin- or α -hederin-treated cells with the presence of CC. This confirmed the involvement of the AMPK-mTOR signalling pathway.

2.8. Hederagenin and α -hederin induce autophagy and degradation of mutant huntingtin proteins via ATG7 gene

Until now, 32 autophagy-related genes (ATG) that are responsible for 3 major mammalian autophagy induction processes including 1) the membrane initiation, 2) nucleation, and 3) elongation and completion in the formation of autophagosome have been identified [48]. Among them, ATG 7 gene is an E1-like enzyme in the 2 major ubiquitin-like protein conjugation systems responsible for the formation of autophagosome. In the first ubiquitin-conjugation system containing ATG12-ATG5-ATG16L, the ATG7 activates the ATG12 through an ATP-dependent manner; in the second ubiquitin-like protein conjugation system, the modification of mammalian homolog of ATG8 (LC3) by the phospholipid phosphatidylethanolamine (PE) responsible for the formation of autophagosomes, is facilitated by the ATG7 [49].

As revealed by fluorescent microscopic analysis in Fig. 8a, both hederagenin and α -hederin significantly increases the GFP-LC3-II puncta formation in ATG7-wild type (ATG7+/+), but not in ATG7-knockout (ATG7-/-) mouse embryonic fibroblasts (MEFs), which is resistant to autophagy induction [50]. Western blots results further showed that both compounds induce a significant increased protein expression of LC3-II in ATG7+/+, but not in ATG7-/- MEFs (Fig. 8b). The results confirmed the ATG7-dependent autophagic properties of hederagenin and α -hederin.



Components	Retention time (min)	Concentration (μM) and (%) in 250 μg/ml fraction		
		HH-NF(AHS)-EF	HH-NF	HH-EF
α-hederin	7.125	22.62 (6.79%)	0.42 (0.13%)	19.88 (5.96%)
Hederagenin	9.221	74.42 (14.05%)	0.38 (0.07%)	2.95 (0.56%)

Fig. 4. Identification of the autophagic components presented in HH (EF) and HH-NF(AHS)-EF.

(a) The total ion chromatogram (TIC) of HH (NF), HH (EF) and HH (AHS)-EF. (b) The TIC of the cell membrane chromatography (CMC) samples of HH (EF) after incubating with PC-12 cells. (c) The structures and mass spectrums (MS) of the two main identified components, hederagenin and α-hederin, which possessed binding affinity toward PC12 cell membrane. (d) The exact concentration (μM) and the percentage (%) of hederagenin and α-hederin in 250 μg/mL of HH (NF), HH (EF) or HH-NF(AHS)-EF were summarized in the table. a: The extract ion chromatogram (EIC) of hederagenin of HH (EF); b: The EIC of hederagenin of HH (NF); c: The EIC of hederagenin of HH (AHS)-EF; d: The EIC of hederagenin in the standard solution; e: The EIC of α-hederin of HH (EF); f: The EIC of α-hederin of HH (NF); g: The EIC of α-hederin of HH (AHS)-EF; h: The EIC of α-hederin in the standard solution.

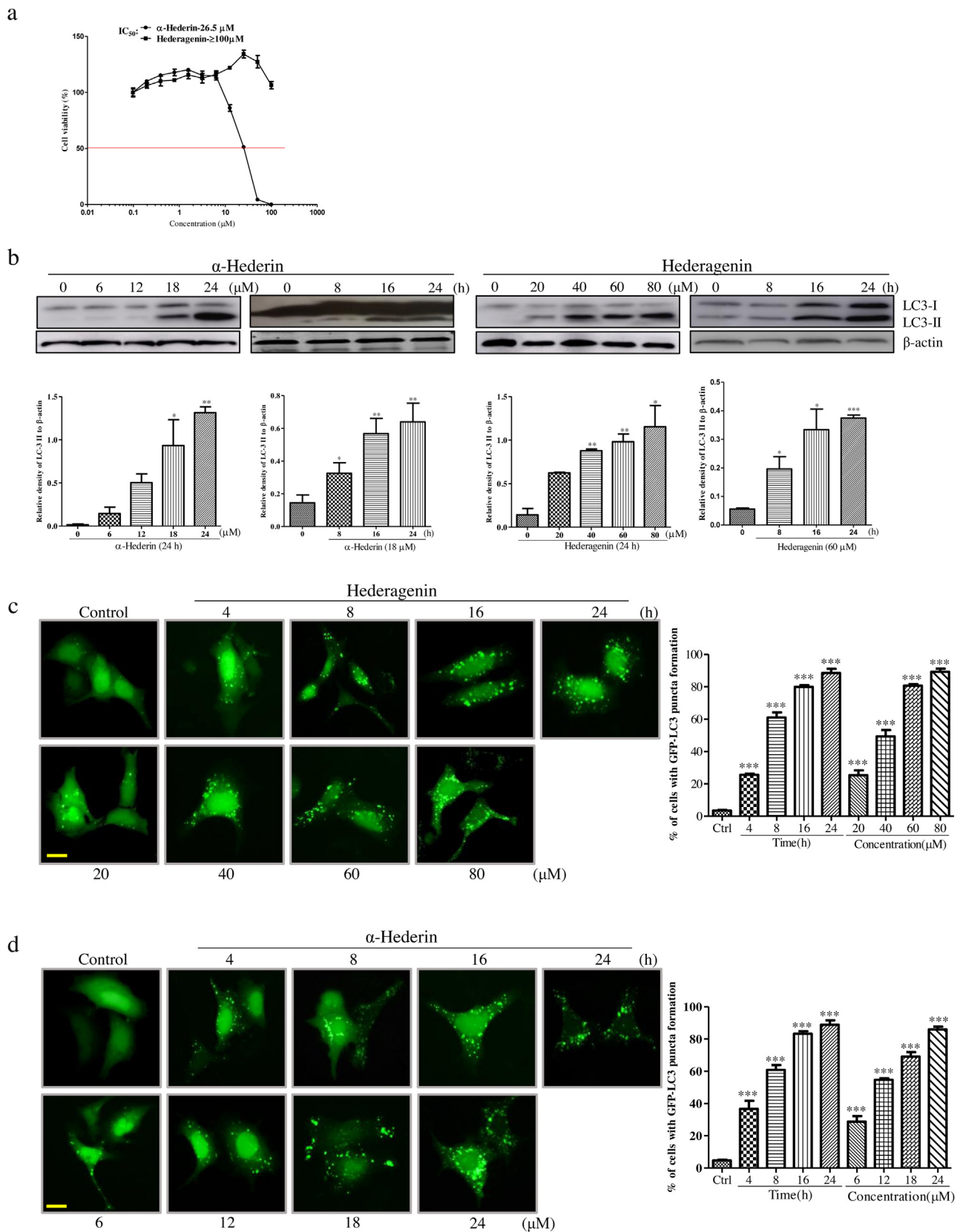


Fig. 5. Cytotoxicity and autophagic effect of hederagenin and α -hederin.

(a) IC_{50} values of the hederagenin and α -hederin were evaluated by MTT assay after 48 h of treatments. (b) PC-12 cells were treated with hederagenin or α -hederin at the indicated time and concentrations. Cells were then harvested for analysis of LC3 I/II with band intensities quantified and normalized to β -actin. (c) PC-12 cells transfected with GFP-LC3 plasmids were treated with hederagenin, or (d) α -hederin at the indicated time and concentrations. Representative fluorescence images with the formation of GFP-LC3 puncta were captured. Right: percentage of cells with GFP-LC3 puncta formation. Bars are representatives of three independent experiments. Error bars, S.D. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Magnification, $\times 40$; Scale bar: $15 \mu m$. The full-length blots are presented in Supplementary Fig. S6.

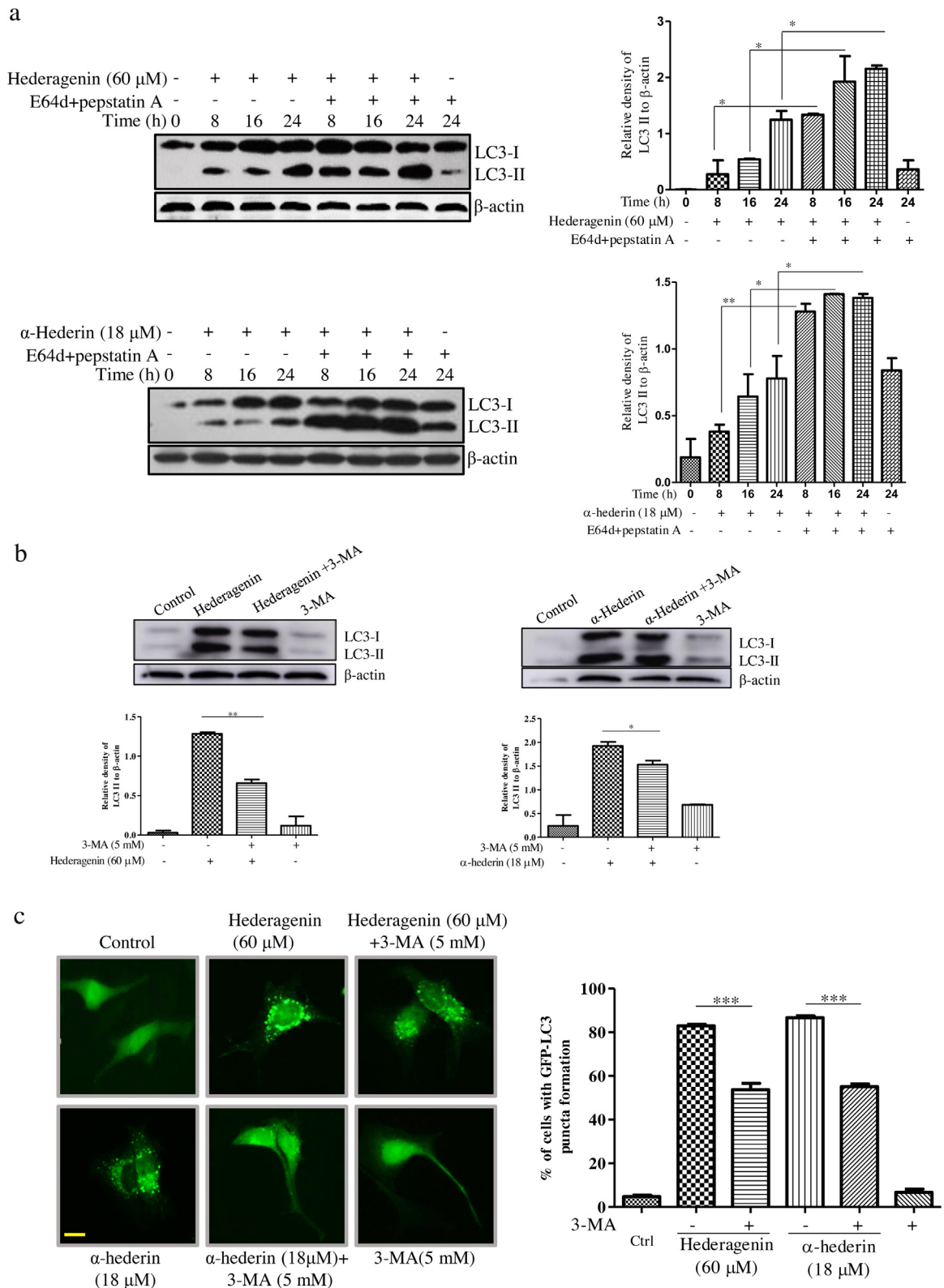


Fig. 6. Hederagenin and α -hederin induces autophagic flux in PC-12 cells.

(a) PC-12 cells were treated with hederagenin (60 μ M) (upper panel) or α -hederin (18 μ M) (lower panel) with or without the presence of lysosomal protease inhibitors (10 μ g/mL) for 24 h. Cells were then harvested for analysis of LC3 I/II with band intensities quantified and normalized to β -actin. (b) PC-12 cells were treated with hederagenin (60 μ M) or α -hederin (18 μ M) with or without autophagic inhibitor, 3-MA (5 mM), for 24 h. Cells were then harvested for analysis of LC3 I/II with band intensities quantified and normalized to β -actin. (c) PC-12 cells transiently transfected with the GFP-LC3 plasmid were treated with hederagenin (60 μ M) or α -hederin (18 μ M) with or without 3-MA (5 mM), for 24 h. Representative images of cells with GFP-LC3 puncta formation and bar charts with percentage of autophagic cells were shown. Bars are representatives of three independent experiments. Error bars, S.D. *** p < 0.001; ** p < 0.01; * p < 0.05. Magnification, $\times 40$; Scale bar: 15 μ m. The full-length blots are presented in Supplementary Fig. S7.

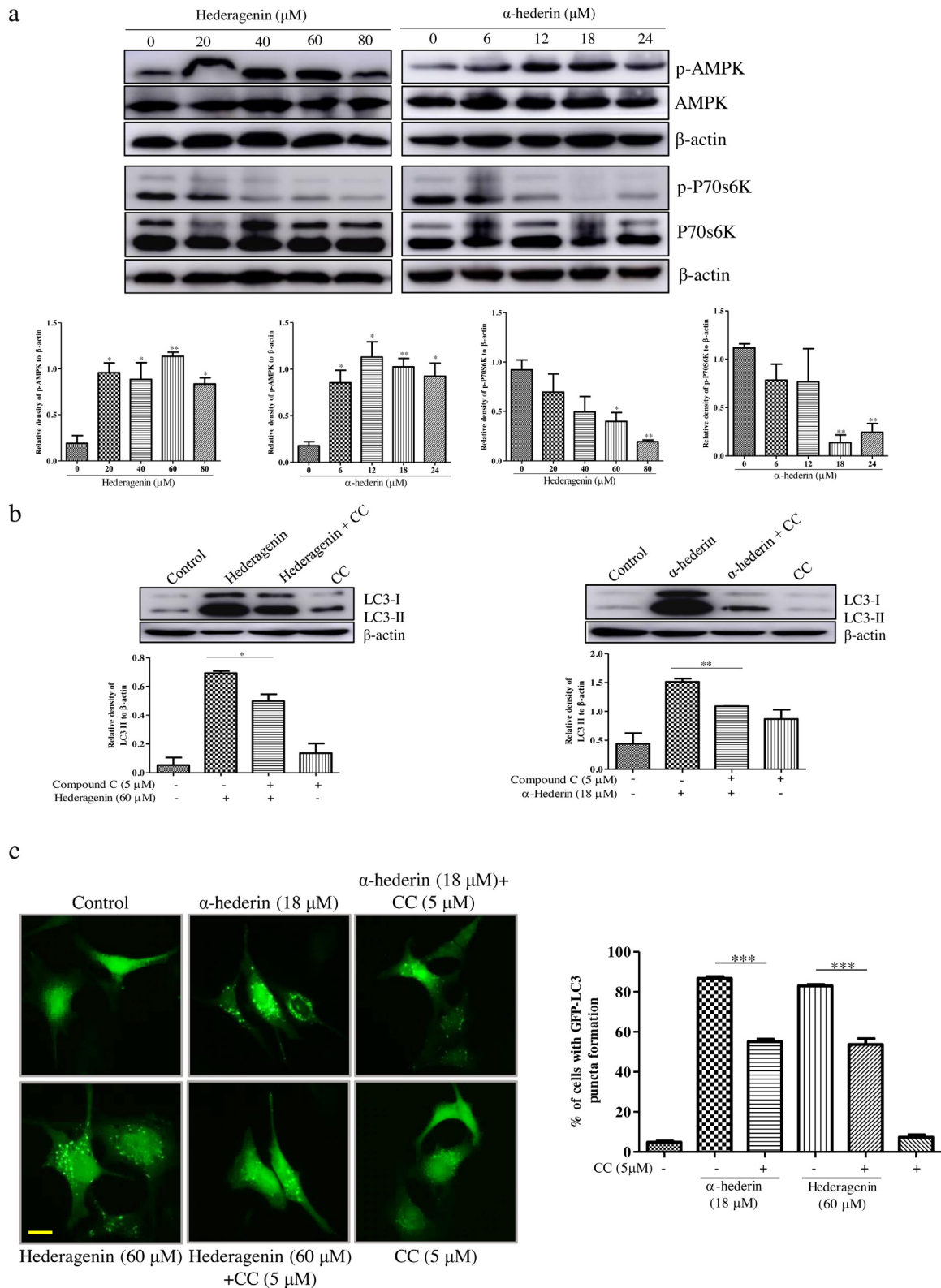


Fig. 7. The activation of AMPK-mTOR signaling pathway by hederagenin and α -hederin.

(a) PC-12 cells were treated with hederagenin (0–80 μM) or α -hederin (0–24 μM) for 24 h. Cells were then harvested for analysis of p-AMPK, total AMPK, p-p70S6K and total p70S6K. Band intensities were quantified by normalization to β -actin. (b) PC-12 cells were treated with hederagenin (60 μM) or α -hederin (18 μM), with or without AMPK inhibitor compound C (CC, 5 μM) for 24 h. Cells were then harvested for analysis of LC3 I/II with band intensities quantified and normalized to β -actin. (c) PC-12 cells were transfected transiently with the GFP-LC3 plasmid for 24 h. Cells were then treated with hederagenin (60 μM) or α -hederin (18 μM), with or without 5 μM of the CC for a further 24 h. The cells were then fixed and analyzed by fluorescence microscopy; Quantitation on the percentage of autophagic cells with GFP-LC3 puncta formation were showed in bar chart. Bars are representatives of three independent experiments. Error bars, S.D. *** p < 0.001; ** p < 0.01; * p < 0.05. Magnification, $\times 40$; Scale bar: 15 μm . The full-length blots are presented in Supplementary Fig. S8

The autosomal-dominant neurodegenerative disorder, HD, is caused by a mutation in a single gene, which resulted in a more than 35 number of CAG trinucleotide repeats and finally a long polyglutamine tract (polyQ) located in the N-terminus of the huntingtin protein (HTT). This kind of toxic and aggregate-prone protein are inaccessible to the small barrel of proteasome, and contribute to various human neurodegenerative diseases [51]. However, recent studies have demonstrated the protective role of autophagic degradation of the mutant huntingtin proteins in a variety of models [52,53]. In our study, with the potent autophagic effect induced by hederagenin and α -hederin, we therefore evaluated the potential protective effect of the compounds in facilitating the autophagic degradation of mutant disease proteins in cellular models.

To begin, ATG7-wild type or -knockout MEFs were transiently overexpressed with the EGFP-tagged mutant huntingtin with 74 CAG trinucleotide repeats (EGFP-HTT 74) before compounds treatment [54,55]. As demonstrated in Fig. 8b, both hederagenin and α -hederin facilitate the clearance of overexpressed mutant huntingtin in ATG7+/+ but not ATG7-/- MEFs, suggesting the autophagy (ATG7) dependent degradation of mutant EGFP-HTT 74. Furthermore, a higher percentage of cells with EGFP-mutant huntingtin inclusion formation was found in ATG-/- MEFs when compared with ATG7+/+ MEFs after compounds treatment, suggesting the protective autophagic role on the clearance of mutant huntingtin inclusion in wild type ATG7 MEFs by both hederagenin and α -hederin (Fig. 8c). Consistently, both hederagenin and α -hederin facilitate the clearance of overexpressed huntingtin in PC-12 cellular model transiently overexpressed with the EGFP-HTT 74 (Fig. 8d). These results suggested the potential neuro-protective role of the compounds in their future therapeutic application.

2.9. Hederagenin and α -hederin facilitate the degradation of mutant A53T α -synuclein (α -syn) in doxycycline (Dox)-inducible cellular model

Degeneration of dopaminergic neurons is one of the common pathological features of PD. This degeneration is often accompanied by the cytoplasmic inclusion formation of Lewy bodies which contain fibrillar aggregates of α -syn. The proto-fibrillization rate of mutated α -syn is higher than its wild-type α -syn protein, while the fibrillation rate of A53T mutant α -syn is higher than the A30P mutant α -syn [56,57]. In fact, A53T mutation in the α -syn gene was found in early onset PD, and mutated forms of α -syn at high concentrations were susceptible to self-aggregation [58]. Since α -Syn aggregation was found in both classical and autosomal dominant early-onset PD [59], through establishing a doxycycline (Dox) - inducible mutant A53T α -syn expression cellular system [60], we examined if hederagenin and α -hederin facilitate the degradation of mutated α -syn *in vitro*. In this cellular model, the overexpression of A53T mutant α -syn was induced by the addition of doxycycline (Dox) [29]. In Fig. 9a, while 1 μ g/mL of Dox induced an overexpression of myc-tagged-A53T- α -syn in cells, both hederagenin and α -hederin facilitate the clearance of the overexpressed mutant α -syn in PC-12 cells after induction of Dox.

In addition to mutation of α -syn, oligomerization of α -syn was reported to contribute to neurotoxicity and pathogenesis of PD by permeating and disrupting the lipid bilayers cellular membranes [61]. To this end, bimolecular fluorescence complementation (BiFC) model, which quantitate the degree of oligomerization of α -syn was set up [62]. In this model, α -syn was fused with either 1) non-fluorescent GFP-N terminal (GFP-N- α -syn), or 2) non-fluorescent GFP-C terminal (GFP-C- α -syn) fragment for *in vitro* quantitation of the oligomerization of α -syn. In general, upon the oligomerization of α -syn, the 2 incomplete non-fluorescent N-terminal-GFP and C-terminal-GFP fragments reconstitute into complete green fluorophore, and give positive GFP fluorescent signal which could

be quantitated by flow analysis. As revealed by a decrease in the fluorescence intensity showed in Fig. 9b, both hederagenin and α -hederin inhibit the oligomerization of α -syn in the BiFC assay, with hederagenin more potent in the inhibition when compared to α -hederin, suggesting the protective role of the 2 active components from HH in modulation of PD.

2.10. Hederagenin and α -hederin rescue cells from cell death induced by MPTP

To begin, the protective effect of hederagenin and α -hederin in cytotoxicity cellular model induced by MPTP was investigated. Firstly, the percentage of cell viability under MPTP treatments (0–2 mM) was determined by MTT assay (Fig. 10a). PC-12 cells were then treated with 0.5 mM of MPTP to induce toxicity in cells [63]. As shown in Fig. 10a and b, while MPTP induces cell death in PC-12 cells, the addition of hederagenin or α -hederin rescue cells from MPTP-induced cell death as revealed by an increased in cell density (Fig. S1), MTT assay (Fig. 10b) and flow cytometry analysis (Fig. 10c). Furthermore, without obvious cytotoxicity induced by hederagenin or α -hederin at its effective concentration of 80 μ M and 24 μ M, respectively (Fig. 10b), we suggested the potential therapeutic use of both autophagic compounds as neuroprotective agents.

3. Discussion

Both hederagenin and α -hederin are isolated from the leaves of *Hedera helix*, a traditional herbal plant prescribed for treatment of upper respiratory tract diseases [64]. While recent literature has reported the apoptotic property of hederagenin in human colon cancer cells [65], α -hederin was reported for its effects in regulating the β 2-adrenergic receptor [64] which has essential role in relaxation of airway, modulating allergy and inflammation via inhibition on the release of proinflammatory mediators [66], and possess protective effect on lung inflammation [67]. Other reported biological roles of α -hederin included anti-virus, anti-oxidation, anti-leishmanial and anti-spasmodic activities [14]. In this study, we reported for the first time the autophagic activity of both hederagenin and α -hederin in PC-12 cells. By demonstrating their abilities in facilitating the degradation of neurodegenerative disease proteins in cellular models, and improving the motor deficits of neuro-toxin injected mice, we therefore proposed the potential neuro-protective role of both hederagenin and α -hederin.

With the recent pathophysiological study on α -syn, small molecules that can inhibit or alter the aggregation of intrinsically disordered proteins have been identified. For example, antibiotic rifampicin [68] and flavonoid baicalein inhibits fibrillation of α -syn [69]. Polyphenol (–)-epi-gallocatechin gallate (EGCG) [70], curcumin [71], and anti-PD drug selengiline [72] were also demonstrated to mitigate the toxicity of α -syn aggregates. The alleviation of cytotoxicity is proposed to be achieved via remodeling of toxic aggregated proteins into non-toxic oligomeric aggregates [61]. In our previous studies, we have demonstrated that small molecule neferine and specific saponins [33,73] isolated from herbal plants, are novel autophagic inducers which facilitate autophagic degradation, and alleviate toxicity of the neurodegenerative disease proteins. Our observation is correlated to the proposed therapeutic role of autophagy in degrading toxic protein aggregates, recycling non-essential components for new protein synthesis, and mitigating toxicity via inhibition of cell death [74], and therefore confirmed autophagy as an alternative pathway to modulate neurodegenerative disorders.

According to the hypothesis on the formation of protein aggregates, initiation of protein aggregation starts with the covalent

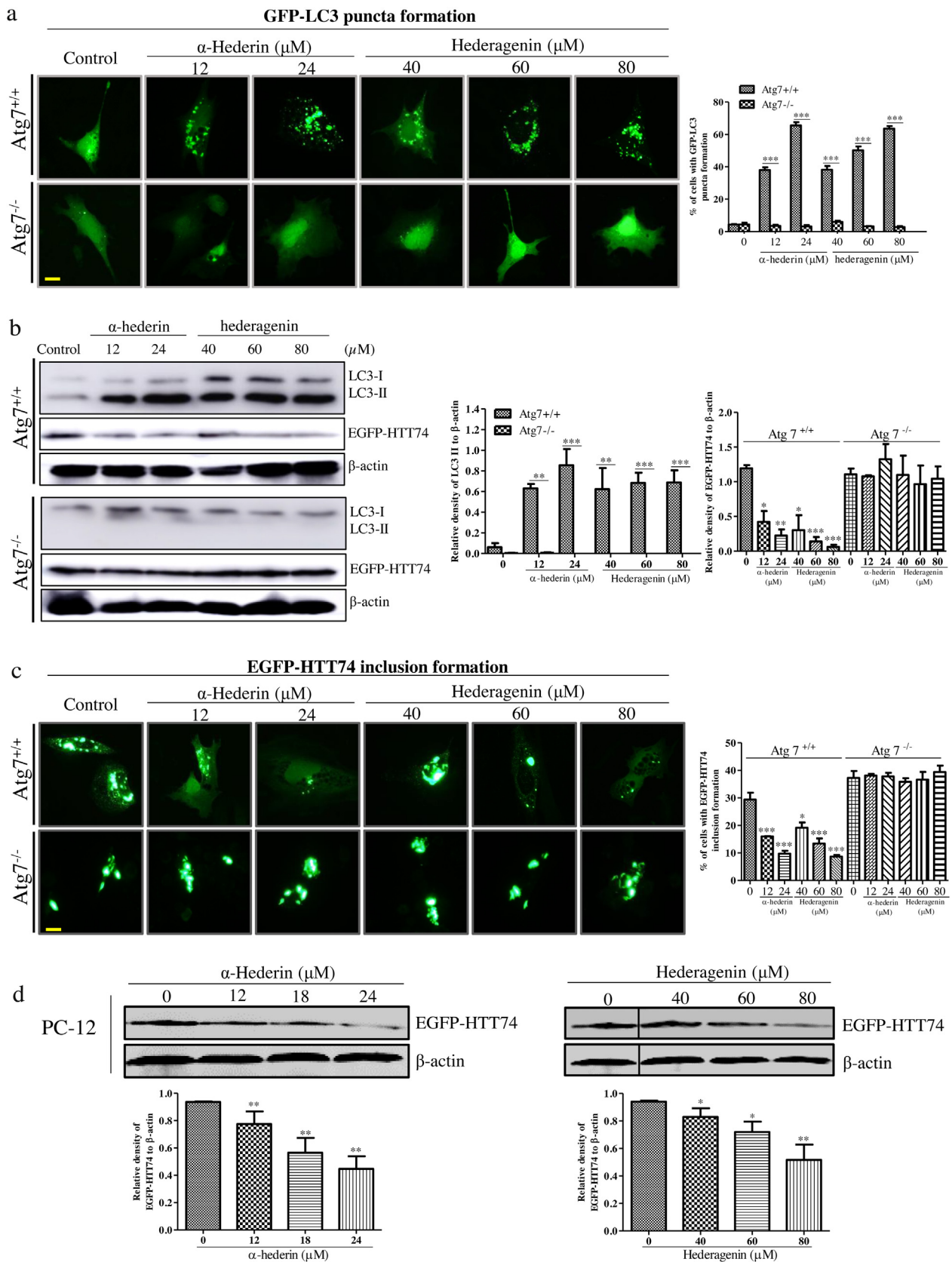


Fig. 8. Hederagenin and α -hederin induces autophagy and HTT inclusion clearance via an ATG7-dependent mechanism.

(a) Both wild type ATG7 and ATG7-deficient MEFs transfected with GFP-LC3 were treated with hederagenin or α -hederin at the indicated concentrations for 24 h. Representative fluorescence pictures of the cells with GFP-LC3 puncta formation were captured. Bar chart indicated the percentage of cells with GFP-LC3 puncta formation. (b) Wild type ATG7 and ATG7-deficient MEFs overexpressed with EGFP-HTT 74 were treated with hederagenin (0–80 μ M) or α -hederin (0–24 μ M) for 24 h. Cell lysates were then harvested and analyzed for LC3 I/II, EGFP and β -actin, respectively. Right: band intensities were quantified and normalized to β -actin. (c) Wild type ATG7 and ATG7-deficient MEFs transfected with EGFP-HTT 74 were treated with hederagenin or α -hederin at the indicated concentrations for 24 h before fluorescent microscopy analysis. Representative

modification of normal proteins to abnormal proteins, followed by the formation of oligomeric intermediates and protofibrillar structures [75]. Through the association of these protofibrillar intermediates, amyloid fibers will form and finally resulting in the formation of aggregates or inclusions of proteins. Of note, it was suggested the intermediate proteins are more toxic than the precursor or the inclusions of the aggregated proteins, and inhibition at the early step of the formation of aggregates would be beneficial [75]. In the current study, we have therefore demonstrated the role of both hederagenin and α -hederin in facilitating the removal of mutant α -syn and huntingtin, and inhibiting the oligomerization of α -syn in cells, confirmed their role in the removal of neurotoxic oligomeric proteins or intermediates in cells.

Ideally, both hederagenin and α -hederin are able to exert their protective biological functions with effective concentrations that possess no obvious cytotoxicity. Consistently, the HH-fraction does not possess observable toxicity to mice as reflected by the normal body weight throughout the experiments (data not shown), suggesting the potential therapeutic role of the two compounds. By using DHBA as a standard control, further measurement on dopamine (DA) level was performed in both the normal and MPTP-treated groups by using UHPLC-MS/MS system, which contains the Agilent 1290 Infinity UHPLC and Agilent 6460 Triple Quadrupole MS system equipped with an electrospray ionization interface used to generate positive ions $[M+H]^+$ (Fig. S2a,b). Our results demonstrated that injection of MPTP lead to a significant increase of dopamine level in mice brain, and the administration of HH-fraction did not further alter the dopamine level of MPTP-treated mice after treatments. Both real time PCR and Western blot analysis (Fig. S3a,b) demonstrated a decrease in the level of α -syn in MPTP-treated mice after HH treatments. The results proposed that the protective function of HH may exert through facilitating the degradation of α -syn in mice. As shown in our results, although there was a rescue in the behavioral deficits in MPTP-treated mice after HH treatments, there is no obvious change in the protein level of α -syn before and after MPTP injection (Fig. S3a,b). This may be explained by the fact that, although MPP⁺ ion can lead to a rapid decrease in ATP level in the brain with neurotoxicity, the hydropyridine or its metabolite was not detected in the brain after 12 h [76]. Besides, the decrease in the ATP level did not persist after 24 h of MPTP injection [77], suggesting the actual neuronal degeneration phenotype may develop at a later stage after MPTP injection [78], and explained for the similar protein level of α -syn in control and MPTP-treated mice. In addition, although MPP⁺ is a substrate for the dopamine transporter (DAT) [37], it can also trigger the release of dopamine [79], which is consistent with our findings that mice showed an increase of dopamine level in brain after MPTP injection, while treatment of HH fraction did not show a further increase in dopamine level of MPTP-mice as showed in Fig. S2.

Although neurotoxins are commonly used to set up PD animal models via the depletion of dopamine, which finally lead to motor deficits in primates *drosophila* and mice, there are limitations in the models. For example, MPTP failed to induce the degeneration of dopaminergic neurons in rats [59], suggesting the genetic factors contributing to the variation on the effect of MPTP in triggering depletion of dopamine, loss of dopaminergic neurons and motor deficits in different animal species. This variation is suggested to be dependent on the activity of MAOB, the enzyme that is responsible for the conversion of MPTP to its toxic species MPP⁺ [37]. With high MAOB activity in brain than in the liver [80], the most sensitive

species to MPTP is C57BL/6 mice [81], which was therefore adopted for setting up the PD animal model in our study. However, special cautions are required when interpreting the behavioral data of rodent's models, because locomotion is controlled differently among species, which lead to variation in the duration of motor deficits after administration of MPTP [82]. For example, while there was literature reporting the long lasting motor deficit in mice after toxin administration [83,84], there were reports depicting the restoration of motor function [85]. It is also noted worthy that there were literatures reporting no changed or hyperactive locomotion in mice after MPTP injection [37,86].

In summary, by depicting the protective role of autophagy in facilitating the degradation of neurodegenerative disease proteins such as α -syn and huntingtin, our study has provided novel insight into the drug discovery of autophagic inducers from traditional Chinese medicinal plants, and suggested the possible molecular mechanisms and therapeutic role of the newly identified active components, hederagenin and α -hederin, from HH in modulating PD or HD.

4. Conclusion

Although there is a long history of prescribing HH as an herbal medicine in Chinese community, the exact molecular mechanisms and the active compounds of HH remains unclear. In our current study, we have identified both hederagenin and α -hederin, as novel autophagic inducers from HH. By applying cell membrane chromatography, we have confirmed the binding abilities of hederagenin and α -hederin toward cellular membrane of PC-12 cells. Based on their abilities in facilitating the degradation of mutant proteins including huntingtin and α -syn via autophagy, we have proposed the neuro-protective effect of both hederagenin and α -hederin in modulating the PD and HD. Neuro-toxin (MPTP) injected mice model has further demonstrated the potential protective role of HH in neuro-therapy.

Neurons, without the induction of cell division, are affected by the accumulation of unwanted cellular organelles or waste burdens throughout their lifetime. Therefore, it was reported that neurons are depending on autophagy for clearance of autophagic substrates [87], and malfunction of autophagy lead to accumulation of ubiquitinated protein aggregates and degeneration of neurons [88]. Based on the fact that brains were found affected in primary lysosomal disorders, and also genes responsible for autophagic and endosomal pathway were found mutated in neurodegenerative disorders, therefore, the protective role of autophagy for maintaining healthy neurons is proposed [74]. Taken together, our current work may provide novel insights into the neuroprotective properties, mechanisms and potential therapeutic values of both hederagenin and α -hederin, isolated from herbal plant HH, via the induction of autophagy.

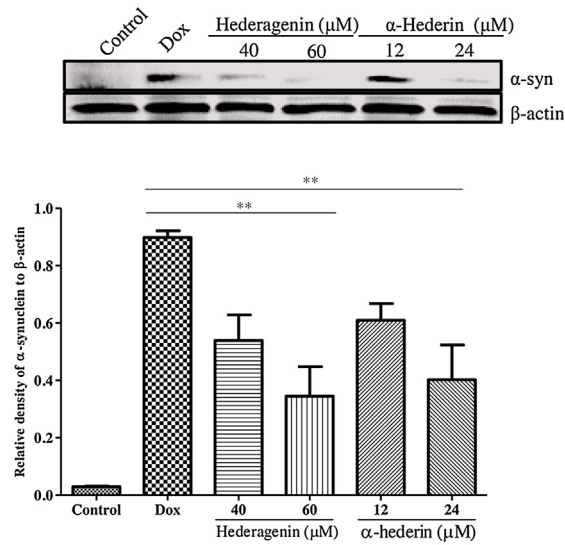
5. Methods

5.1. Antibodies and reagents

The plant HH was purchased from Luo Shen Xing Zhong Yao Cai Co. Ltd. (Hebei, China), hederagenin and α -hederin were obtained from Nanjing Spring & Autumn Biological Engineering Co., Ltd. (Nanjing, China). All the chemicals and reagents used in the study were purchased from Sigma-Aldrich (MO, USA) unless

pictures showed the formation of EGFP-HTT 74 inclusions. Bar chart indicated the percentage of cells with GFP inclusions formation under different treatment conditions. (d) PC-12 cells transfected with EGFP-HTT 74 were treated with hederagenin or α -hederin at the indicated concentrations for 24 h. Cell lysates were then harvested and analyzed for EGFP and β -actin, respectively. Black line represents chopped image in the same blot under same exposure. Band intensities were quantified and normalized to β -actin. Bars are representatives of three independent experiments. Error bars, S.D. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Magnification, $\times 40$; Scale bar: 15 μ m. The full-length blots are presented in Supplementary Fig. S9.

a



b

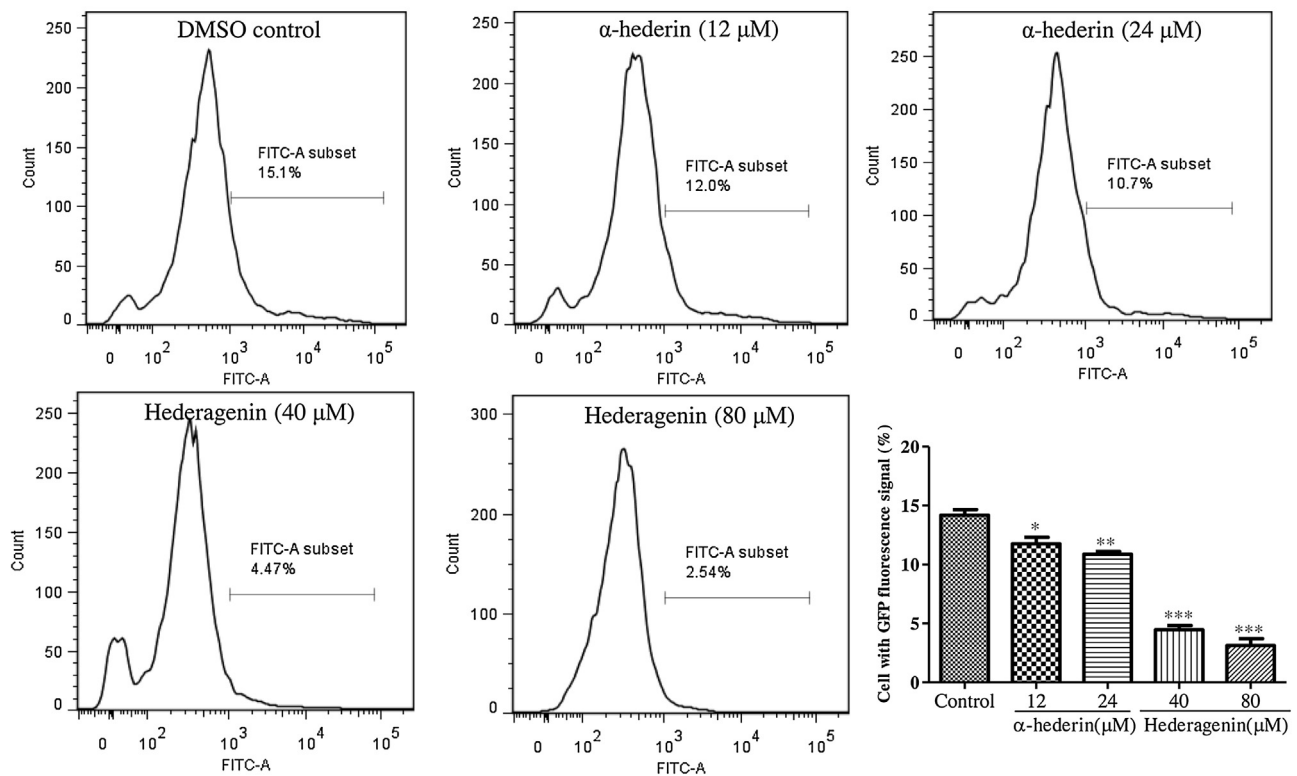


Fig. 9. Hederagenin and α-hederin facilitates the clearance of mutant α-synuclein.

(a) Doxycycline inducible PC-12 cells transfected with A53T α-syn plasmids were induced for the overexpression of mutant A53T α-syn by doxycycline (1 μg/mL). The expression of mutant A53T α-syn was then switched off by the removal of doxycycline after 24 h. The cells were then incubated with hederagenin or α-hederin for a further 24 h. The protein level of myc-tagged-mutant α-syn was measured by immunoblotting with antibody against myc-tag. Band intensities were quantified and normalized to β-actin. (b) HeLa cells transfected with both non-fluorescent GFP-N terminal (GFP-N-α-syn) and non-fluorescent GFP-C terminal (GFP-C-α-syn) fragments were incubated with hederagenin or α-hederin for 24 h. The percentage of cells with oligomerization of α-syn that gave out fluorescence signal, was measured by flow cytometry. Bar chart indicated the percentage of cells that produced GFP-positive signal. Data from the flow cytometry analysis is represented as means ± S.D. of three independent experiments. Bars are representatives of three independent experiments. Error bars, S.D. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. The full-length blots are presented in Supplementary Fig. S10.

otherwise specified. EGFP-LC3 plasmid was gift from Tamotsu Yoshimori (Osaka University, Osaka, Japan). EGFP-HTT 74 plasmid was gift from David C. Rubinsztein (University of Cambridge, Cambridge, UK). Monoclonal antibodies against AMPK (#2532), p-AMPK

(Thr172) (#2535), Myc-tag (9B11) (#2276) or LC3B (#3868s), and polyclonal antibodies against p70S6K (#9202), p-p70S6K (Thr389) (#9205) were obtained from Cell Signaling Technologies Inc. (Beverly, MA, USA). Monoclonal anti-GFP (sc-9996) antibodies were

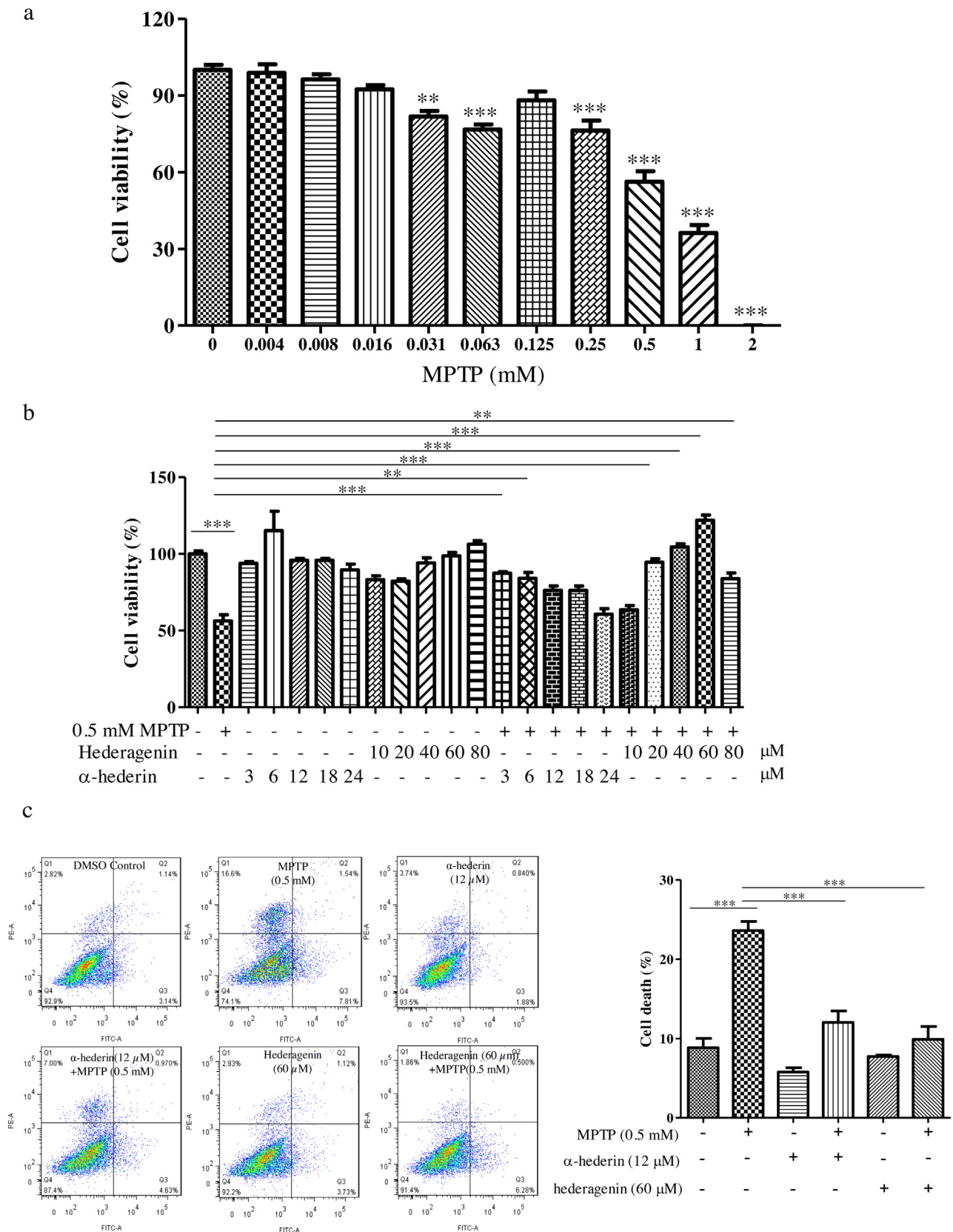


Fig. 10. The protective effect of hederagenin and α -hederin in MPTP-induced cell death.

(a) The cytotoxicity (% of cell viability) of MPTP in PC-12 cells after 48 h of treatment with the indicated concentrations. (b) Cell viability of PC-12 cells treated with hederagenin (10–80 μ M) or α -hederin (3–24 μ M) in the presence or absence of 0.5 mM of MPTP. Cell viability was measured by MTT assay after 48 h of treatment. (c) Cell viability of PC-12 cells treated with hederagenin (60 μ M) or α -hederin (12 μ M) in the presence or absence of 0.5 mM of MPTP. Cell viability was measured by flow analysis after propidium iodide staining for cell death. Data from the flow cytometry analysis is represented as means $\pm$ S.D. of three independent experiments. Bars, S.D. *** p < 0.001; ** p < 0.01; * p < 0.05.

obtained from Santa Cruz Biotechnology (Santa Cruz, CA, USA). Monoclonal β -actin antibodies (A2228) were purchased from Sigma (St. Louis, MO, USA).

5.2. The identification of the bioactive components in HH (EF) and HH-NF(ASH)-EF

The identification of the bioactive components in HH (EF) and HH-NF(ASH)-EF were performed by using CMC method [33]. The measurement of the bioactive components, hederagenin and α -hederin in HH (EF), HH (NF) and HH-NF(ASH)-EF were carried out by UHPLC (Agilent Technologies 1290 Series) equipped with the time of flight MS (Agilent Technologies 6230) with a jet stream ion source, which was operated in negative ion mode during the analysis. All the samples were separated and analyzed on an Agilent Zorbax Eclipse Plus C-18 column with a particle size of 1.8 μ m (flow rate: 0.35 mL/min). The mobile phase was set as follow: mobile phase A (0.1% formic acid in water) and mobile phase B (0.1% formic acid in ACN): 0–8 min, 5–70% B; 8–11 min, 70–100% B; 11–14 min, 100% B; 14.1–18 min, 5% B. For UHPLC-TOF/MS analysis, the data was acquired in the scan mode (m/z 100 to 1700 Da with 2.0 spectra/s). Data were analyzed by using Agilent MassHunter Workstation software B.01.03.

5.3. Cell culture

All cells were purchased from the American Type Culture Collection (ATCC) (Rockville, MD, USA), unless otherwise specified. PC-12 cells cultured with DMDM medium were supplemented with 10% horse serum, 50 μ g/mL streptomycin and 50 U/mL penicillin (Invitrogen, Paisley, Scotland, UK). ATG7-wild-type and ATG7-deficient mouse embryonic fibroblasts (MEF) provided by Masaaki Komatsu (Juntendo University, School of Medicine, Tokyo, Japan), were supplemented in DMEM with 10% fetal bovine serum. HeLa cells were maintained in MEM. All cells were cultured in a humidified incubator at 37 °C with 5% CO₂ unless otherwise specified.

5.4. Protein extraction and Western blots

Cells were harvested by RIPA lysis buffer (Cell Signaling Technologies, Beverly, MA, USA), and with the final protein concentrations measured by using Bradford reagent (Bio-Rad, Hercules, CA, USA). For biochemical analysis of the mice tissue, the whole mouse brain was homogenized in 1 mL cold PBS solution by using the TissueRuptor® and TissueLyser Distruption systems (Qiagen). Homogenized tissues were then lysed by RIPA buffer before measurement of protein concentration. After electrophoresis for protein band separation, proteins were transferred to the nitrocellulose membrane which was blocked with 5% non-fat dried milk immediately after protein transfer. Corresponding primary antibodies, followed by the addition of HRP-conjugated secondary antibodies, were incubated with the protein membrane with constant shaking. Visualization of protein band pattern was performed by using the ECL Western Blotting Detection Reagents (Invitrogen, Paisley, Scotland). Band intensities were quantified by using the software, ImageJ (ImageJ 1.46r; National Institutes of Health, Bethesda, MD, USA).

5.5. Real time PCR analysis

PC-12 cells transfected with the EGFP-HTT 74 plasmids were incubated with different concentrations of hederagenin or α -hederin for 24 h. Total RNA was extracted by using the FavorPrep™ Total RNA purification mini kit (Favorgen, PingTung, Taiwan). SuperScript® VILO™ Master Mix (Invitrogen, Scotland, UK) was used for the synthesis of cDNA via reverse transcription. SYBR®

Select Master Mix (ThermoFisher Scientific, Waltham, MA, USA) was used for targeted gene detection according to the manufacturer's instructions. Primers 5'- AGG CAG CTG GAA AGA CAA AA-3' and 5'- CAG CTC CCT CCA CTG TCT TC-3' were used to detect α -syn in mice brain. All of the real-time PCRs were carried out on the ViiA™ 7 Real Time PCR System (Applied Biosystems, Grand Island, NY, USA).

5.6. Immunocytochemistry and fluorescence microscopy

In brief, GFP-LC3 puncta formation was analyzed as described previously in [29]. Firstly, cells were plated on top of coverslips inside a 6-well culture dish. After compounds treatments, cells were fixed with 4% paraformaldehyde for 20 min. FluorSave™ mounting media (Calbiochem, San Diego, CA, USA) was used to mount the coverslips with cells before subjected to fluorescence microscopic analysis. The number of GFP-positive cells, and cells with GFP-LC3 puncta formation were examined and counted under the Nikon ECLIPSE 80i microscope by using 40 $\times$ of magnification. In order to standardize the quantitation, we defined the percentage of cells with autophagy induction by counting the number of cells with punctate GFP-LC3 fluorescence in GFP-positive cells. A minimum of 1000 GFP-positive cells from 3 randomly selected fields were scored.

5.7. Cytotoxicity assays and flow cytometry

Cell viability was measured by using the MTT method (3-[4,5-dimethylthiazol-2-yl]-2,5-diphenyl tetrazolium bromide) as described previously in [29]. Absorbance (OD) of cell samples was obtained by spectrophotometer at 570 nm. The percentage of cell viability was calculated by using the formula: cell viability (%) = cells number_{treated}/cells number_{DMSO control} $\times$ 100. All MTT data were calculated from three independent experiments. Cell viability was also measured by flow cytometry using the annexin V staining kit (BD Biosciences, San Jose, CA, USA). In brief, PC-12 cells treated with hederagenin and α -hederin were analyzed by multiparametric flow cytometry using FITC-annexin V and/or propidium iodide staining (BD Biosciences, San Jose, CA, USA) by following manufacturer's instructions. Flow cytometry was then carried out using a FACSCalibur flow cytometer (BD Biosciences). Data acquisition and analysis were performed with CellQuest (BD Biosciences).

5.8. Detection of mutant huntingtin proteins or inclusions

To further investigate the protective effects of hederagenin and α -hederin, PC-12 cells were transfected with GFP-HTT 74 plasmids by using Lipofectamine Plus LTX reagent (Invitrogen, Paisley, Scotland, UK). Cells were then treated with either hederagenin or α -hederin for a further 24 h to determine the protein level of mutant huntingtin by Western blot analysis using antibody against GFP [55]. On the other hand, ATG7 wild-type and ATG7-deficient MEF cells transfected with EGFP-HTT 74 were analyzed for the formation of huntingtin inclusion after 24 h of hederagenin or α -hederin treatment.

Percentages of cells with cytoplasmic GFP-huntingtin (GFP-HTT74) inclusion formation were counted by the number of cells with GFP inclusions over the total number of GFP-positive cells in the same field under fluorescent microscopic analysis. More than 200 GFP-positive cells were scored for each treatment.

5.9. Bimolecular fluorescence complementation (BiFC) assay

Both (1) non-fluorescent GFP-N terminal (GFP-N- α -syn), or (2) non-fluorescent GFP-C terminal (GFP-C- α -syn) plasmids were generous gifts from Pamela J. McLean (Department of Neuroscience,

Mayo Clinic Florida, Jacksonville, FL, USA). HeLa cells transfected with both GFP-N- α -syn and GFP-C- α -syn plasmids were incubated at 37 °C for 4 h. Cells were then subjected to different concentration of hederagenin or α -hederin treatments for a further 24 h in a humidified incubator at 30 °C [62]. Green fluorescent signal which indicates the reconstitution of complete GFP proteins, was detected by flow analysis (BD FACSAria III, San Jose, CA, USA).

5.10. Detection of mutant α -synuclein in doxycyclin-inducible PC-12 cells

Firstly, doxycyclin-inducible PC-12 cells were transfected with A53T mutant α -synuclein plasmid, a generous gift from C.B. KO (The Hong Kong Polytechnic University, Hong Kong, China) for 24 h. Transfected cells were then induced for the overexpression of mutant myc-tag A53T α -synuclein proteins by adding 1 μ g/mL of dox. To switch off the overexpression of α -synuclein, dox-containing medium was removed from PC-12 cells before compounds treatment [60]. Cells were then treated with hederagenin or α -hederin for 24 h. The protein level of mutant myc-tag α -synuclein was analyzed by immunoblotting against myc antibody.

5.11. Preparation of HH (EF) and HH-NF(ASH)-EF fractions for animals study

500 g of HH powder was first immersed in 5 L of 75% ethanol for 1 h, and then refluxed twice with 75% ethanol for 1 h, respectively. The extracted solution was combined and dried by rotary evaporation at 60 °C under vacuum condition. The dried extract was re-dissolved in water and then partitioned with petroleum ether (1:1 vol/vol) for 3 times to obtain HH-PF, the remained water fraction was then partitioned with ethylethanoate (1:1 vol/vol) for 3 times. All the extracted ethylethanoate solution (HH-EF) was combined and dried. The remaining water layer was then further partitioned with *n*-butanol (1:1 vol/vol) for 3 times and dried to give the HH-NF and HH-WF, respectively. Finally, WF, NF, EF, PF fraction of HH were obtained for further UHPLC-TOF/MS and biological activity analysis.

For the acid hydrolysis of HH-NF, 20 g HH-NF was re-dissolved in 200 mL of water, then hydrochloric acid was added into the HH (NF) solution until the pH 2.5. This acid solution was heated for 2 h at 100 °C. After cool down, the acid solution was partitioned with ethylethanoate (1:1 vol/vol) for 3 times to give the HH-NF(ASH)-WF and HH-NF(ASH)-EF, which were dried for the further UHPLC-TOF/MS and biological activity analysis.

5.12. MPTP neurotoxin- induced motor deficits mice model

8-week-old C57BL/6 mice (female body weight: 22 $\pm$ 4 g and male body weight 25 $\pm$ 4 g) were purchased from The Chinese University of Hong Kong (Hong Kong). The animals were housed in a 12 h light/dark cycles and temperature-controlled room and given ad libitum access to food and water. All of the experiments were carried out in accordance to the "Institutional Animal Care and User Committee guidelines" of the Macau University of Science and Technology. A mixture of HH (EF) and HH-NF (ASH)-EF in 1:1 ratio was prepared by dissolving in Mill-Q water. C57BL/6 mice were randomly divided into four experimental groups as follows: (1) Normal control group without treatments; (2) MPTP-treated group; (3) MPTP-treated mice with 40 mg/kg HH extract treated; (4) MPTP-treated mice with 80 mg/kg HH extract treated. To begin, all HH (EF) extract treatment groups were gavage-fed with 40 mg/kg or 80 mg/kg of HH extract daily for 10 consecutive days before neuro-toxin injection (pre-treatment). Neurotoxin MPTP (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine) was dissolved in buffer saline, and given to mice at a daily dose of 20 mg/kg/day by

intraperitoneal (i.p.) injection [89] for 10 consecutive days with or without HH extract feeding. Oral feeding of HH extract was performed 2 h before the MPTP injection.

5.12.1. Behavioral swimming test

Animals were subjected to a swimming test 30 min after MPTP administration. Mice were observed by independent examiners who were blinded to the different treatments between groups. Each animal was placed in a container of water which the depth was about 10 cm. The water temperature was controlled at about 22–25 °C. Swimming score was recorded with modification that used by Marshall and Berrios [90]. In general, the scoring criteria were as follows: Mice with continuous swimming movement: 3; Mice with occasional floating: 2.5; Mice with floating >50% of time: 2.0; Mice with occasional swimming only: 1.5; Mice with occasional swimming using hind limbs while floating on side: 1.0. Mice with no use of limbs: 0. Each animal was scored at 1-min intervals for 10 mins [91].

5.12.2. Behavioral rotarod test

The rotarod test, which requires animals to balance and walk on a rotating cylinder, is a widely used test to evaluate motor coordination and motor learning of the MPTP-mice [92,93]. To begin, mice were placed in a balanced position on a stationary 2 cm diameter cylinder at 30 min after MPTP injection. In 10 s time, the rotarod was accelerated to a full speed of 20 rpm, then motor coordination can be assessed by comparing the latency to fall over time between treatment groups. Each mouse was evaluated for 3 times (with a maximum cut-off time of 1200 s/time). To ensure accuracy of the obtained results, mice have to be trained to run on the rod at a higher speed than their normal pace before experiments started [94,95].

5.13. Preparation of mice brain tissues for measurement of DA, hederagenin and α -hederin

Whole mice brains were collected and weighted after sacrificed. Each brain tissue was put into 500 μ L of NaCl solution and ultra-sonicated for 40 s to give the whole tissue homogenate. Each tube of homogenate was set to a final volume of 800 μ L. To begin, 200 μ L of each tube of brain homogenate was spiked with 10 μ L of 3, 4-dihydroxybenzylamine (DHBA) (0.5 mg/mL). 200 μ L of methanol was then added to precipitate proteins with vortexing and centrifugation at 10000 rpm for 10 min. While the supernatant was collected, and remaining precipitation was re-extracted with 200 μ L of methanol. The collected supernatant was then combined and dried by using dry nitrogen blowing instrument. The residue was reconstituted with 200 μ L of 70% acetonitrile, and then 10 μ L of the aliquot was injected into the UHPLC-MS/MS system [96].

5 mg of DA were dissolved with methanol to give a final stock standard solution of 0.5 mg/mL and stored at –80 °C. For preparing the standard curve of DA, serial dilution on DA standard solution combined with 10 μ L of DHBA (0.5 mg/mL) was performed. 5 mg of hederagenin or α -hederin were dissolved in DMSO to give a stock solution of 50 mM. This stock solution was further diluted with methanol to a final concentration of 50 μ M before serial dilution on hederagenin and α -hederin were prepared for the standard curve.

5.14. The measurement of dopamine, hederagenin and α -hederin level in mice brain tissues

Internal standard (IS, DHBA), DA, hederagenin and α -hederin level in the mouse brain, were quantified using UHPLC-MS/MS system which includes Agilent 1290 Infinity UHPLC, and Agilent 6460

Triple Quadrupole, equipped with an electrospray ionization interface used to generate positive ions for the determination of DA; negative ion mode for the measurement of hederagenin and α -hederin in mouse brains. The compounds in plasma samples were analyzed by using the Agilent Zorbax Eclipse Plus C-18 column with a particle size of 1.8 μ m (flow rate: 0.35 mL/min). The mobile phase for the determination of DA was 98% mobile phase A (0.1% formic acid in water) and 2% mobile phase B (0.1% formic acid in ACN) in 5 min. The mobile phase for the measurement of hederagenin and α -hederin was set as follows: mobile phase A (0.1% formic acid in water) and mobile phase B (0.1% formic acid in ACN): 0–5 min, 5–50% B; 5–7 min, 50–95% B; 7–10 min, 95% B; 10–11 min, 95–5% B; 11–15 min, 5% B. The column and auto-sampler temperature were maintained at 40 °C and 4 °C, respectively. Data were analyzed by using Agilent MassHunter Workstation software B.01.03.

The gas temperature was set at 325 °C with a flow rate of 10 L/min. Gases were set at 40 psi for the nebulizer, capillary, 4000 V. The fragmentor was 90, 90, 110, 135 and 300 for DA, DHBA, α -hederin and hederagenin, respectively. The collision energy was set at 3, 1, 55 and 46 for DA, DHBA, α -hederin and hederagenin, respectively. The mass transitions were as follows based on multiple reaction monitoring: m/z 154.0 $\rightarrow$ 137.0 for DA, 140.1 $\rightarrow$ 123.1 for DHBA, 795.3 $\rightarrow$ 471.4 for α -hederin and 471.3 $\rightarrow$ 393.1 for hederagenin, respectively. Quantification was performed by selected reaction monitoring of the product ion for DA, by using the IS method with peak area ratios and a linear least-squares regression curve. The measurements of α -hederin and hederagenin in mouse brain were done by using the standard and linear least-squares regression curve.

5.15. Statistical analysis

All results were expressed as means $\pm$ SD. The difference was considered statistically significant when the p -value < 0.05 . Either Student's t -test or one-way ANOVA analysis was applied for comparison among different groups in current study.

Competing interests

The authors declare that they have no competing interest.

Authors' contributions

A.G.W., V.K.W.W. and B.Y.K.L. designed, carried out the experiments, analyzed the data and prepared the draft of manuscript. Z.W. performed the animal experiments, data analysis and interpretation of behavioral studies. Y.Z.Z. and A.C.Y.L. conceived the idea and revised the manuscript. L.L. and B.Y.K.L. conceived the idea, supervised all research and revised the manuscript. All authors reviewed the manuscript.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.phrs.2016.11.002>.

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